Leveraging NI 43-101 Reports within a LLM Structure

March 19, 2026 – V001

March 20, 2026 – V002

Project Thesis

- The mining industry is large and highly regulated, but its complexity can make it difficult to act upon specific opportunities and trends. This complexity and the efforts to improve transparency for public investors led to the establishment of certain disclosures.
- Specifically, the NI 43-101 is a technical disclosure protocol that can be triggered by a number of conditions, including technical ones related to a specific project. These documents are authored by Qualified Professionals and undergo a certain level of technical review by QPs employed by the exchanges.
- These documents are highly structured and publicly available, and as such provide an excellent repository of similarly written but distinct material, with content dedicated to one prospect / ore deposit / mine per report.
- This provides a consistent starting point for a ML / LLM approach for extracting knowledge and value from these data

Proposed Workflow

- Ingest NI 43-101 Form, Companion Policy, and prior SEDAR-filed NI 43101 Technical Reports
- Categorize NI43101 Reports into the following:
    A. Property of Merit / Early Stage Technical Report
    B. Maiden Mineral Resource Estimates
    C. PEA
    D. PFS
    E. FS
    F. Other Material Technical Changes
    G. Non-technical triggers (e.g., Mergers, Takeovers, First Time Reporting, etc…)
- Develop database initially with additional categories driven by expert rules, such as Payable Metals and Geography, and then allow the system to develop other possible patterns, such as Proximity to Grid Power, Access Complexity, Process Mineralogy, Multiple Exchange-listed, Reports with …
- All categories can receive a certain baseline level of analysis, and then certain categories will receive further questions based on the types of questions pertinent to the project. For example, there is little need to go into comminution technology opportunities for a Property of Merit. Similarly, identifying a potential need for MMI surveys is not applicable to a project with a FS. Thus, a decision tree branching structure can be imagined

Early Products

Pre-Publication Assessments

- Mining company: Pre-submission Document Building Service
  - Develop predictive framework for expediting report writing, pulling from pre-defined and suggested analogous 43101 reports
  - Identify project risks and opportunities, as compared to prior reports and projects
  - Suggest additional technical work that could improve the quality of the report
  - Suggest additional technical work that could improve the quality of the project
- Mining company: Pre-submission Review Service
  - Compare a "completed" but unreviewed document to prior accepted reports, find deficiencies, suggest improvements to the quality of the report → leverages the Document Building Service
- Exchange: Technical Review Tool
  - Compare the document prior accepted reports → give it a "score"
  - Find technical deficiencies, provide "checklist" for reviewer(s)

Published Report Assessments

- Mining Company: Gap Analysis
  - Compare project to adjacents and provide prediction of opportunities. For example, Orogenic Gold Project A could benefit from the learnings of Orogenic Gold Project B that is +10 years along the development sequence which encountered delays here and there
  - Upgrading of Mill A to HPGR yielded great results, do testwork there before SAG tests… or places with missed improvements on CAPEX/OPEX
- Investment Companies: Project Status Opportunities
  - Flagging of newly submitted 43101, comparison to past projects, and predict a timeline to production (or failure). Chance to purchase stocks or invest capital into the projects most likely to succeed, or alternatively, provide a heads up to existing investors that the project might be a dud and they should back out.
    - E.g., a good fit for Precious Metal Streaming or strategic investment at the 'right time'
  - Identification of under-appreciated critical metal opportunities / mineralization
- Government
  - Trend analsys, risk asssement
- Consulting Companies: Services Fit Opportunities
  - WSP Golder requests Gap Analyses broadly of recent projects or a specific project to create new contracts.
  - Reach out to consulting exploration groups (e.g., Anomalous Exploration) and provide a workplan they can take to filing-company X
- Exploration Company: Mineralization trend expansion
  - Identify missed opportunities stated by the filing company for a recent report, and act upon that, such as through property acquisition
  - Re-evaluate older reports for opportunities not taken or dropped, that are now available in the current landscape
- Established Technology Companies: Technical Fit Opportunities
  - Tomra SBS XRT POS well suited to FS details of Project X
- Emerging Technology Companies: Technical Fit Opportunities
  - Automation and phasing out of diesel mobile equipment would likely lead to X % improvement in performance metrics
  - Utilization of Muon Tomogrpahy or Shovel Mounted Sensing would enable additional disocveries or reduce dilution in the ore body

Probable increasing complexity

		REPORT STAGE / AGE			
		Pre-Submission	Submission	Published	Superseded (old / not current)
CUSTOMER TYPE	Mining Co	Build, Review		Gap Analysis	Retrospective
	Exchange		Reviewing Agent		
	Investment Co			Y	Y
	Consulting			Y	Y
	Established Tech			Y	Y
	Emerging Tech			New Clients	Y
	Govt			Y	Y
	Exploration Co			Y	Y

Needs building out!!

Input Data Sets

- Principal (Structured and Predictable, Authored by QP's)
  - NI43101 Reports
- Adjacent (Semi structured and sometimes raw data)
  - Metal prices
  - Assessment Reports
  - Annual reports on activities (e.g., BCGS)
  - Theses, academic papers, textbooks, reference books
  - News Releases, especially if QP-guaranteed and/or structured
  - Data rooms
- Distal (Unstructured, variable age and focus)
  - News articles
  - conference abstracts
  - Maps
  - Production reports
  - Data Rooms



Example NI 43-101 Reports from SEDAR+

	Profile(s)	Document	Submitted date	Principal jurisdiction	File size	Actions
☑	Alamos Gold Inc. / Alamos Gold Inc. (000037377)	Technical report (NI 43-101) - English.pdf	20 Mar 2026 11:07 EDT	Ontario	11188 KB	Generate URL
☑	Yukon Metals Corp. / Yukon Metals Corp. (000055282)	Technical report (NI 43-101) - English.pdf	19 Mar 2026 21:03 EDT	British Columbia	48784 KB	Generate URL
☑	Lithium Argentina AG (000027103)	Technical report (NI 43-101) - English.pdf	19 Mar 2026 17:12 EDT	British Columbia	27039 KB	Generate URL
☑	Purecore Metals Inc. (000111452)	Technical report (NI 43-101) - English.pdf	19 Mar 2026 14:35 EDT	British Columbia	5416 KB	Generate URL
☑	Edge Copper Corporation / Edge Copper Corporation (000032534)	Technical report (NI 43-101) - English.pdf	18 Mar 2026 18:35 EDT	British Columbia	16623 KB	Generate URL
☑	Orla Mining Ltd. / Orla Mining Ltd. (000025619)	Technical report (NI 43-101) - English.pdf	18 Mar 2026 17:37 EDT	British Columbia	21444 KB	Generate URL
☑	RPX Gold Inc. (000004007)	Technical report (NI 43-101) - English.pdf	18 Mar 2026 16:54 EDT	Ontario	23526 KB	Generate URL
☑	Clinch Resources Ltd. / Clinch Resources Ltd. (000052069)	Technical report (NI 43-101) - English.pdf	17 Mar 2026 22:11 EDT	British Columbia	26688 KB	Generate URL

Data Counts (# 43-101 Reports per day)

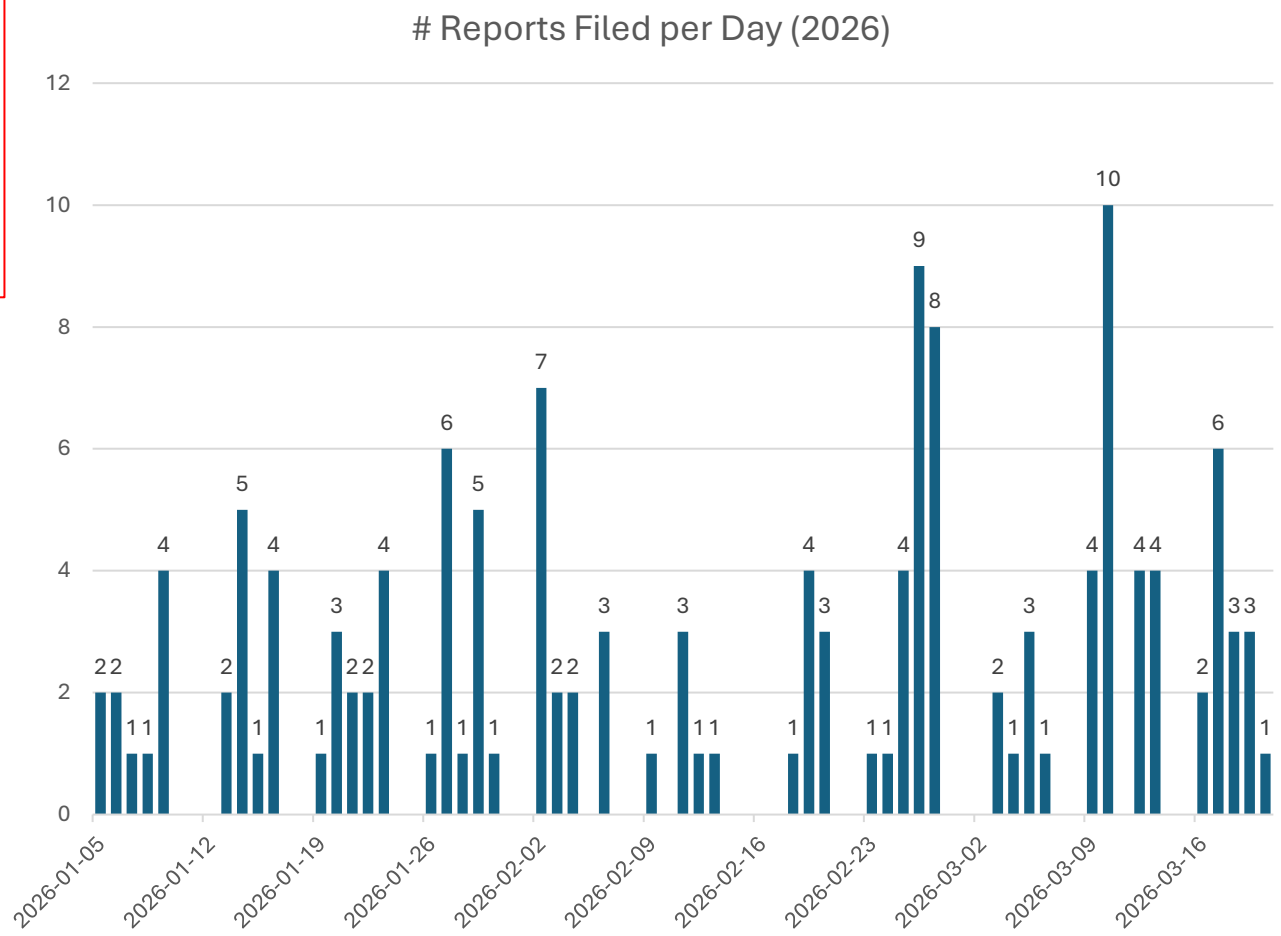
Based on a 2019 report on NI 43-101 disclosure basics, the TSX (Toronto Stock Exchange) had **1** dedicated technical review staff member, while the TSX Venture Exchange (TSXV) had **3**. OSC.ca +1

These individuals are part of a broader group of technical reviewers across Canadian jurisdictions (including the BCSC, OSC, and AMF) who monitor compliance with NI 43-101 standards for over 1,300 listed mining companies. Professional Geoscientists Ontario (PGO)

<u>2026</u>

Jan	48
Feb	51
Mar (to 20th)	44

Reviewers are likely overloaded



Basic Prompts into ChatGPT provide outputs

📊 Final Cleaned Classification Summary (All 30)

Property / Early Stage

- Yukon Metals
- Purecore
- Bighorn
- Leading Edge
- Greenridge
- Alicanto
- American Salars
- United Lithium

Maiden Mineral Resource Estimate

- Metallic Minerals *(confirmed MRE)*

PEA

- Edge Copper
- RPX Gold
- Canadian Energy Metals
- Gold X2
- New Found Gold *(PEA-supported filing)*

PFS

- Thor Explorations

FS

- Orla Mining
- Alamos Gold

Other Material Technical Changes

- Lithium Argentina
- Sayona
- Minera Alamos
- Ur-Energy
- Mako Mining
- Thunder Gold

Non-Technical Triggers

- Clinch Resources
- Ventra Metals
- EU Gold Mining (x2)
- Auranova
- Centenario Gold
- 1548043 B.C. Ltd.

Example Use Case – Pre-Submission and Exchange Reviews

- The 30 most recent reports contained a range of categories. One example showed a report that contained small deficiencies visible during a cursory review and should have been flagged and the report undergo revisions:
  - Significant mismatch between budgeted dollars in Summary and Recommendations
  - Incorrect chapter ordering
  - Table of contents with coordinate systems
  - Maps without north arrows

☑	1548043 B.C. Ltd. (000111221)	Technical report (NI 43-101) - English.pdf	12 Mar 2026 15:16 EDT	British Columbia	8635 KB	Generate URL

8. Exploration 37
Soil Sampling 37
Rock Samples 38
Structural Geology Review 38
Results 38
708000 710000 712000 714000 716000 718000 40
9. Drilling 45
10. Sample Preparation, Analyses and Security 46
Ultrafine Soil Samples 46
Rock Samples 47

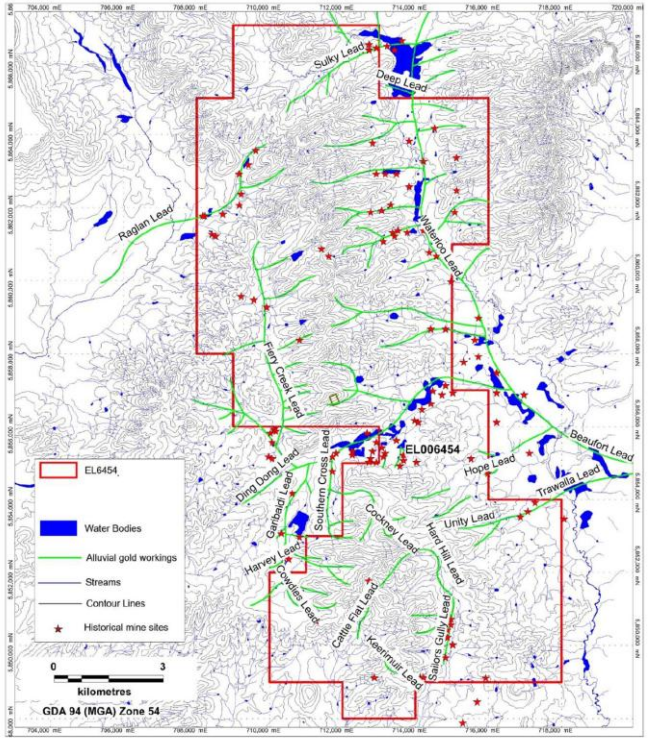


Figure 6: Location of alluvial leads in EL006454.

Microsoft Copilot Inputs and Main Outputs

1. Make a chart showing the main types of formal disclosure documents for publicly listed mining companies, such as a NI 43101
2. Create a timeline of standard developments
3. What are the triggers or purposes for filing a 43101
4. Create a flowchart of filing triggers
5. Create a decision-tree version
6. Can you access recently filed 43101 reports? Also, is there typically just one author? Or more? Which triggers are usually with more than 1 author?
7. Give me a chart that shows the typical progression of a mining project, from prospecting discovery through to mid life, and then indicate when 43101 reports might show up, and which type
8. When a 43101 report is filed, is it categorized neatly into one of the trigger categories?

NI 43-101 Purpose and Triggers

The central purposes of NI 43-101 include:

- Ensuring that public disclosure is accurate, complete, and prepared by a Qualified Person (QP) [pgo.ca]
- Supporting investor decision-making with standardized, reliable technical information [pgo.ca]
- Providing a current, comprehensive summary of material technical information for a material property [pgo.ca]
- Enhancing transparency and comparability across issuers and jurisdictions (as enforced through Form 43-101F1 and NI 43-101 rules) [mrmr.cim.org]

Main Triggers:

- 1. Property Milestone Triggers
- 2. Company Milestone Triggers
- 3. Listing-Related Triggers
- 4. M&A Transaction Triggers
- 5. Unplanned or Compliance-Related Triggers

1. Property Milestone Triggers
These are events related directly to a mineral project. Filing is required upon first disclosure of any of the following for a *material* property:
- **First disclosure of a Mineral Resource estimate** [pgo.ca]
- **First disclosure of a Preliminary Economic Assessment (PEA)** [pgo.ca]
- **First disclosure of a Mineral Reserve estimate** (PFS/FS level) [pgo.ca]
- **Material change to any of the above** (new results, updated estimates, new economic studies) [pgo.ca]

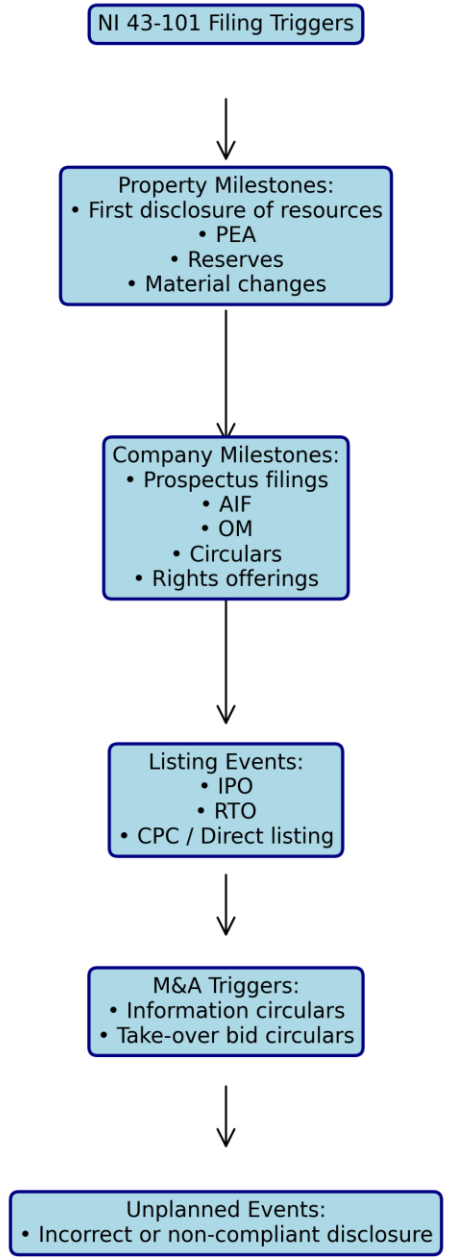
2. Company Milestone Triggers
These apply when the *issuer* undertakes certain corporate actions:
A. Becoming a Reporting Issuer in Canada
•Filing is required **upon becoming a reporting issuer** (e.g., IPO, RTO, CPC listing).
The technical report must cover all *material* mineral properties.
[mrmr.cim.org]
B. Filing Certain Disclosure Documents
A technical report must accompany scientific and technical disclosure in documents such as:
•**Preliminary (long-form) prospectus**
•**Short-form prospectus** (if first-time disclosure or material changes exist)
•**Information or proxy circulars**
•**Offering memorandum**
•**Rights offering circular**
•**Annual Information Form (AIF)**
•**Valuation reports**
•**TSX Venture offering documents**
•**Take-over bid circular** (when disclosing material scientific/technical information)

3. Listing-Related Triggers
A technical report is required when a company attempts to list on a Canadian exchange and must demonstrate that its material properties meet listing requirements.
This includes:
- **IPO**
- **Reverse takeover (RTO)**
- **Capital Pool Company (CPC) transaction**
- **Direct listing**

4. M&A Transaction Triggers
A technical report may be required when scientific and technical information is disclosed in:
- **Information circulars** (for shareholder approval of acquisitions involving securities issuance)
- **Take-over bid circulars** (when the bidder discloses resources/reserves/PEAs on material properties)
- **Any document filed during M&A that contains technical disclosure on a property that will be material to the resulting issuer**

5. Unplanned or Compliance-Related Triggers
Regulators may demand a technical report when:
- **Incorrect, incomplete, or non-compliant disclosure is made**, such as:
  - Misstating historical resource estimates
  - Providing scientific/technical claims unsupported by a current technical report

Regulators can request immediate filing to correct the public record.


NI 43-101 Filing Triggers
Property Milestones:
• First disclosure of resources
• PEA
• Reserves
• Material changes
Company Milestones:
• Prospectus filings
• AIF
• OM
• Circulars
• Rights offerings
Listing Events:
• IPO
• RTO
• CPC / Direct listing
M&A Triggers:
• Information circulars
• Take-over bid circulars
Unplanned Events:
• Incorrect or non-compliant disclosure

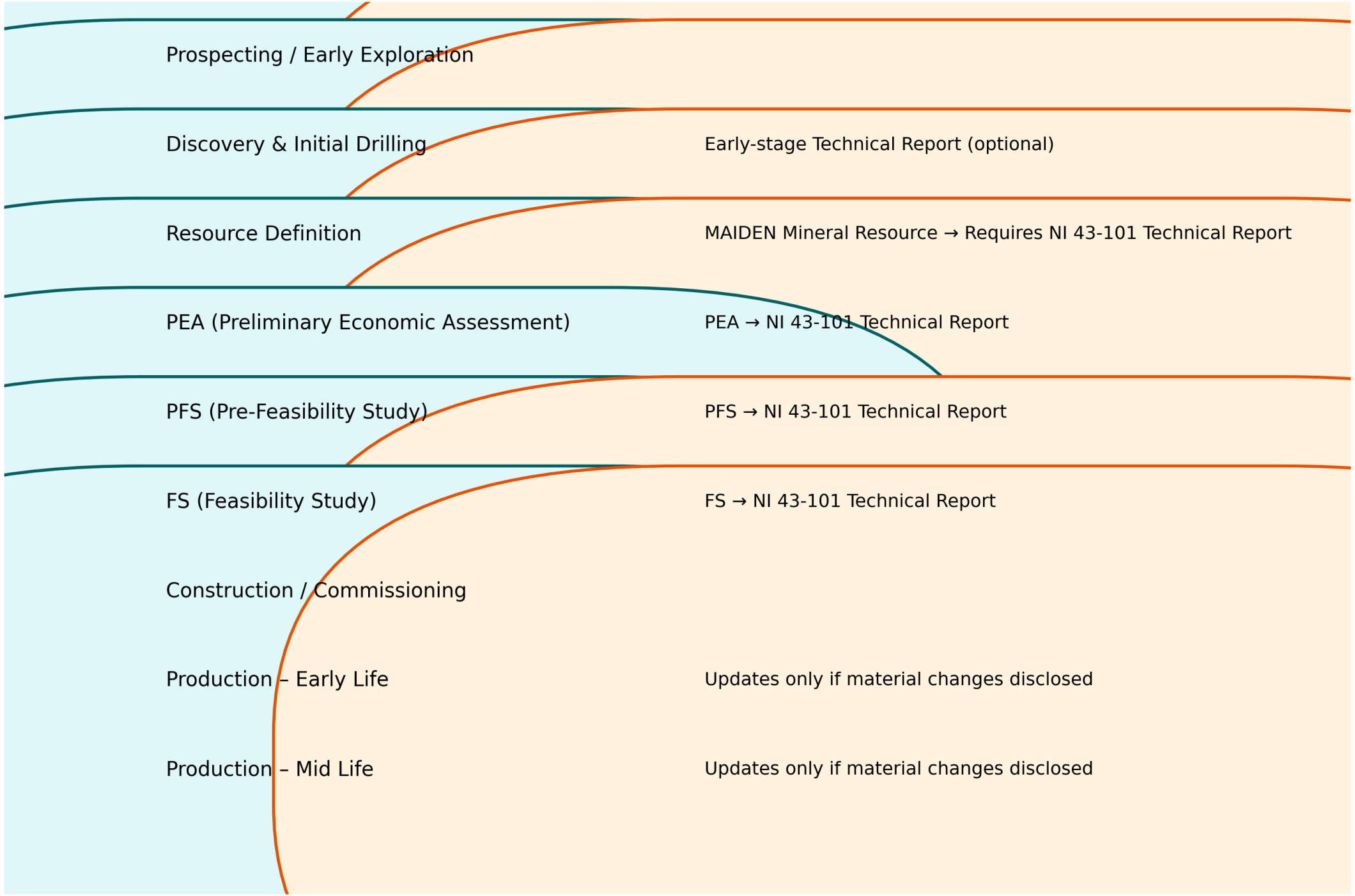
recent examples of NI 43-101 reports filed in 2026 and # of authors

- Yukon Metals filed an independent NI 43-101 report for the AZ Project (effective Dec 4, 2025). It is available on SEDAR+ and the company's website. [taiwannews.com.tw]
  - Yukon Metals' AZ Project report is referred to as an "independent technical report", consistent with QP-led authorship which may be one or many depending on scope. [taiwannews.com.tw]
- Borealis Mining filed an updated NI 43-101 PEA for the Sandman Gold Project, accessible via their news release and SEDAR+. [mining.com]
- Metallic Minerals filed its NI 43-101 technical report for the La Plata Project and its Mineral Resource Estimate (MRE) on SEDAR+. [accessnewswire.com]
  - Metallic Minerals' La Plata Project report was prepared by SLR USA Advisory Inc., which indicates a team-based authorship typical of larger firms. [accessnewswire.com]
- New Found Gold filed the Hammerdown Project PEA NI 43-101 report on SEDAR+ and EDGAR. [miningstoc...cation.com]
  - New Found Gold's Hammerdown Project report was authored by at least 11 Qualified Persons (QPs), all from WSP Canada Inc. [miningstoc...cation.com]
- Thor Explorations filed its NI 43-101 PFS technical report for the Douta Gold Project on SEDAR+. [sharecast.com]

Which types of filings usually have one or more than one author?

- Triggers that usually result in multi-author reports:
- 🟦 Preliminary Economic Assessment (PEA)Includes mine plans, metallurgy, processing, costing → requires multiple QPs.Example: Borealis Mining's updated PEA. [mining.com]
- 🟥 Prefeasibility Study (PFS) or Feasibility Study (FS)Highly detailed engineering-level documents → almost always multi-disciplinary.Example: Thor Explorations PFS report for the Douta Gold Project. [sharecast.com]
- 🟧 Updated Mineral Resource Estimate (MRE)Often involves geology, resource modelling, QA/QC, and sometimes metallurgy.Example: Metallic Minerals' La Plata updated MRE report. [accessnewswire.com]
- 🟪 M&A-triggered reports or listing-triggered reportsThese must provide comprehensive coverage of all technical aspects → usually multi-author.
- 🟫 Large or complex projectsDeep porphyry systems, large gold systems, or critical minerals projects often require many QPs.

- Triggers that more commonly result in single-author reports
- 🟩 These tend to be simpler technical disclosures:Early-stage exploration resultsSimple property descriptionsHistorical data compilationsTriggered corrections of previous non-compliant disclosure (e.g., Patriot Critical Minerals) [sme.asia]One QP with appropriate expertise may be sufficient for these.


Prospecting / Early Exploration
Discovery & Initial Drilling
Early-stage Technical Report (optional)
Resource Definition
MAIDEN Mineral Resource → Requires NI 43-101 Technical Report
PEA (Preliminary Economic Assessment)
PEA → NI 43-101 Technical Report
PFS (Pre-Feasibility Study)
PFS → NI 43-101 Technical Report
FS (Feasibility Study)
FS → NI 43-101 Technical Report
Construction / Commissioning
Production – Early Life
Updates only if material changes disclosed
Production – Mid Life
Updates only if material changes disclosed

- 🔵 Early-Stage (Exploration & Discovery)
  - Reports may appear but are optional unless the company discloses scientific/technical information that triggers NI 43-101 requirements.
- 🟠 Resource Definition
  - MAIDEN Mineral Resource Estimate→ Mandatory NI 43-101 Technical Report (Form 43-101F1)
- 🟣 Economic Studies
  - PEA (Preliminary Economic Assessment) → Requires NI 43-101
  - PFS (Pre-Feasibility Study) → Requires NI 43-101
  - FS (Feasibility Study) → Requires NI 43-101
  - These reports are typically multi-author because they span geology, mining, metallurgy, processing, cost engineering, environmental, etc.
- 🟤 Construction & Production Stages
  - Reports only required if the company discloses material changes (new resources, reserves, updated economics, or major changes to life-of-mine plans).

When a 43101 report is filed, is it categorized neatly into one of the trigger categories?

- No. There is no system on SEDAR+ or in NI 43-101 that classifies filed technical reports into standardized trigger categories.The legislation defines triggers, but the filed reports themselves are not labeled or sorted by these categories in any official taxonomy.
- Even though reports are not categorized, the trigger is usually clear from the content because the NI 43-101 Form (43-101F1) requires the report to state:The type of study (e.g., “Preliminary Economic Assessment…”)The effective dateWhether it supports resource estimates, reserves, PEA, PFS, FS, etc.The specific written disclosure that triggered the filing requirement (a requirement under Part 5). [mrmr.cim.org]This makes it easy to deduce the trigger, even though there is no official “category.”